

BF27.3

Brandon Furman 3/29

Background

Purpose:

What is the relationship between the resistance through a light bulb filament at room temperature and the resistance of the light bulb when the bulb is on?

Hypothesis:

If the light bulb is higher in temperature, then it will have a lower resistance

Variables:

Independent:

- Temperature of the lightbulb

Dependent:

- Resistance of the lightbulb

Constants:

- Lightbulb which is being measured

Justification for hypothesis:

At higher temperatures, the particles in the filament vibrate with more energy. The idea is that this extra vibrations make the electrons move faster when they bump into the filament material, meaning that the electrons move easier and thus there is lower resistance. Another possible reason is that the thermal expansion of the material causes an increase in the length of the resistor, but an squared increase in the area of the resistor, which means a net decrease in resistance by $R = \rho \frac{l}{A}$.

Materials

- A lightbulb
- Two batteries
- Two alligator clips
- A multimeter

Procedure

1. Place the lightbulb and batteries in series using the alligator clips.
2. Use the multimeter to record the voltage through the lightbulb and the resistance of the lightbulb right after connecting the circuit.
3. Record the resistance of the lightbulb at 30 second intervals, stopping at 5 minutes.
4. Use the initial voltage and resistance function to calculate the power dissipated by the lightbulb using $P(t) = \frac{V^2}{R(t)}$. Using $P(t) = \frac{dE(t)}{dt}$, we can derive $E(t) = \int_0^t P(t)dt$ to graph energy dissipated as a function of time.
5. Graph ΔR as a function of E to see the relationship between the amount of energy dissipated and the change in resistance of the lightbulb.

Data and Results Tables

Voltage across lightbulb - V

Time after connecting circuit (s)	Resistance (Ω)
0	
30	
60	
90	
120	
150	
180	
210	
240	
270	
300	